ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

MATHEMATICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 2 hours**

Covers the full Grade 7 year with more problem solving: whole numbers, budgets and loans, exponents, fractions, decimals and percentages, integers, patterns and equations, geometry and constructions, measurement, data handling and probability.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of NINE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL your working. Answers alone may not get full marks.
4. Calculators may NOT be used.
5. Diagrams are NOT necessarily drawn to scale.
6. Write neatly and legibly.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which number is the smallest?

A 0,3 B $\frac{1}{3}$ C 0,305 D 0,25

1.2 The highest common factor (HCF) of 12 and 30 is …

A 2 B 6 C 3 D 60

1.3 15% of 200 is …

A 15 B 20 C 30 D 3

1.4 −5 + 12 − 3 = …

A 4 B −4 C 10 D −10

1.5 $(2 \times 3)^2$ = …

A 12 B 36 C 18 D 10

1.6 Which expression means "5 more than double a number"?

A $5x + 2$ B $2(x + 5)$ C $7x$ D $2x + 5$

1.7 How many lines of symmetry does an equilateral triangle have?

A 3 B 0 C 1 D 2

1.8 A polygon with eight sides is called a …

A hexagon. B heptagon. C octagon. D decagon.

1.9 The volume of a cube with sides of 3 cm is …

A 9 cm^3 B 18 cm^3 C 27 cm^3 D 54 cm^3

1.10 The median of 3 ; 8 ; 5 ; 10 ; 6 is …

A 6 B 5 C 8 D 6,4

[10]

QUESTION 2: WHOLE NUMBERS AND FINANCE

2.1 Is 2 345 divisible by (a) 5 and (b) 3? Give a reason for each answer. (2)

2.2 Use a property of numbers to calculate $56 \times 25 + 44 \times 25$ without long multiplication. (2)

2.3 A school hall has 24 rows with 36 chairs in each row. (a) How many chairs are there? (b) 1 000 people come to a concert. How many people will have to stand? (3)

2.4 The Dlamini family earns R12 000 per month. They spend 30% on rent, R3 600 on food and $\frac{1}{8}$ of their income on transport. Calculate: (a) the amount spent on rent; (b) the amount spent on transport; (c) the amount left after rent, food and transport. (4)

2.5 Mr Khumalo borrows R5 000 at 12% simple interest per year. He pays the loan back after 2 years. Calculate the total amount he must pay back. (3)

[14]

QUESTION 3: EXPONENTS

3.1 Write $2 \times 2 \times 2 \times 2 \times 3 \times 3$ in exponential form. (2)

3.2 Calculate: (a) $5^2 - 2^3$ (b) $\sqrt{(36 + 64)}$ (c) $\sqrt{49} \times 2^3$ (6)

[8]

QUESTION 4: FRACTIONS, DECIMALS AND PERCENTAGES

4.1 Calculate. Give your answers in the simplest form: (a) $\frac{7}{8} - \frac{1}{4} + \frac{1}{2}$ (b) $1\frac{1}{3} \times \frac{3}{8}$ (4)

4.2 Calculate: (a) $2{,}07 \times 1\,000$ (b) $0{,}56 \times 0{,}7$ (3)

4.3 Write $\frac{7}{20}$ as (a) a decimal fraction and (b) a percentage. (2)

4.4 A taxi fare of R250 is increased by 20%. Calculate the new fare. (2)

4.5 Ayanda scored $\frac{3}{5}$ in her first test and 68% in her second test. In which test did she do better? Show your working. (2)

4.6 Arrange from the largest to the smallest: $\frac{2}{3}$; 0,6 ; 65% ; 0,606 (1)

[14]

QUESTION 5: INTEGERS

5.1 Write down the next two numbers in the pattern: -11 ; -7 ; -3 ; … ; … (2)

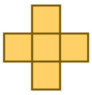
5.2 One winter night the lowest temperatures were: Johannesburg 3 °C, Bloemfontein −2 °C and Sutherland −7 °C. (a) Which town was the coldest? (b) What is the difference between the temperatures of Johannesburg and Sutherland? (3)

5.3 Calculate: $-9 - (-4) + 2$ (1)

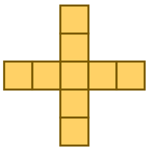
[6]

QUESTION 6: PATTERNS, FUNCTIONS AND ALGEBRA

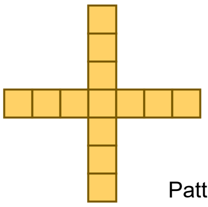
6.1 Tiles are used to make a pattern of crosses.



Pattern 1



Pattern 2



Pattern 3

Pattern number (*n*)	1	2	3	4	10
Number of tiles	5	9	13	(a)	(b)

6.1.1 Write down the values of (a) and (b). (2)

6.1.2 Write down a rule for the number of tiles in Pattern *n*. (2)

6.1.3 Which pattern uses 81 tiles? (2)

6.2 In a flow diagram, each input is multiplied by 3 and then 1 is subtracted. (a) Find the outputs when the inputs are 5 and 7. (b) Which input gives an output of 29? (4)

6.3 Solve for x: (a) $x/3 = 6$ (b) $2x + 7 = 25$ (3)

6.4 Mpho is *m* years old. His sister is 4 years older than he is. (a) Write an expression for his sister's age. (b) Together their ages add up to 30. Write an equation and find Mpho's age. (3)

[16]

QUESTION 7: GEOMETRY

7.1 Describe the steps to bisect a line segment AB using only a compass and a ruler. (3)

7.2 In ΔABC, ∠A = 90° and ∠B = ∠C. Calculate the size of ∠B. (2)

7.3 ΔPQR has sides of 3 cm, 4 cm and 5 cm. ΔXYZ has sides of 6 cm, 8 cm and 10 cm. Are the triangles congruent or similar? Give a reason. (2)

7.4 Name each quadrilateral: (a) "I have four right angles and equal diagonals, but my four sides are not all equal." (b) "All my sides are equal, all my angles are 90° and my diagonals cross at 90°." (2)

7.5 The minute hand of a clock moves from 12 to 3. Through how many degrees does it rotate, and in which direction? (2)

7.6 How many faces does a hexagonal prism have? (1)

[12]

QUESTION 8: MEASUREMENT

8.1 A rectangular vegetable garden is 15 m long and 8 m wide.

8.1.1 Calculate the length of fencing needed to go around the garden. (2)

8.1.2 Fencing costs R85 per metre. Calculate the cost of the fencing. (2)

8.1.3 Calculate the area of the garden. (1)

8.1.4 One bag of compost covers 25 m². How many bags must be bought to cover the whole garden? (2)

8.2 A triangle has an area of 24 cm² and a base of 8 cm. Calculate its perpendicular height. (2)

8.3 A water tank in the shape of a rectangular prism is 2 m long, 1,5 m wide and 1 m high. Calculate its volume in m³ and then in litres. (1 m³ = 1 000 ℓ) (3)

[12]

QUESTION 9: DATA HANDLING AND PROBABILITY

9.1 The stem-and-leaf display shows the test marks (out of 70) of 13 learners.

Stem	Leaves
3	5 8
4	2 2 7
5	0 3 6 6 6
6	1 4 9

Key: 3 | 5 means 35

9.1.1 How many learners scored less than 50? (1)

9.1.2 Write down the mode. (1)

9.1.3 Write down the median. (1)

9.1.4 Calculate the range. (1)

9.2 A spinner has 8 equal sections: 3 red, 2 blue, 2 yellow and 1 green.

9.2.1 What is the probability of landing on red? (1)

9.2.2 What is the probability of NOT landing on blue? Simplify your answer. (2)

9.2.3 The spinner is spun 80 times. About how many times would you expect it to land on red? (1)

[8]

TOTAL: 100

END OF QUESTION PAPER